



Analysis of Boeing 767-300 Airliner

Twinjet Titans

Graydon Carlton, Graham Jones, Dustin Leung, Kevin
Liko



767-300 -300ER

- Airliners
- Wide-body
- Low-Wing
- Twin Engine
- 1986 - Release
- Versions still in production



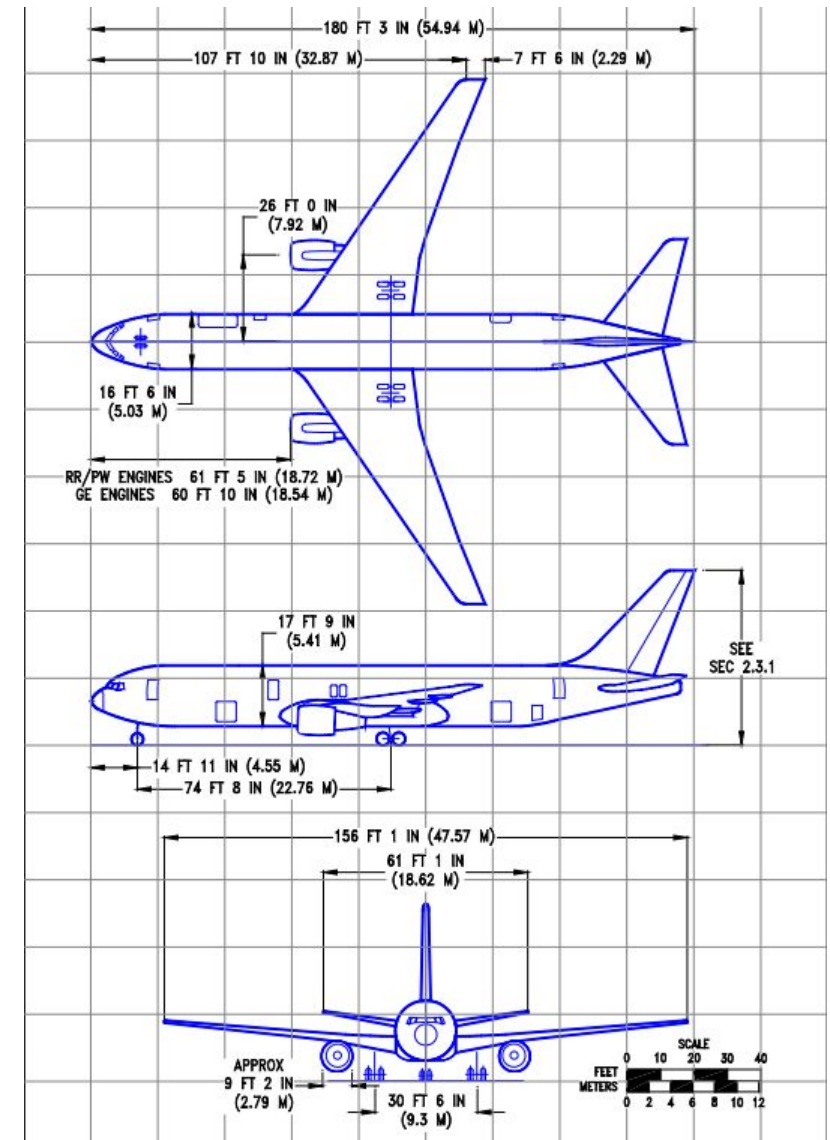
https://en.wikipedia.org/wiki/Boeing_767



Geometric Recreation Results

Wing Results	Value
Wing $c(y)$	$c(y) = 11 \text{ m} - 0.3662y$
Sref Wing	316.1 m^2
Swet Wing	261.4 m^2
$\bar{c}$ Wing	7.5964 m
Xac Wing	8.1683 m
Yac Wing	9.2944 m

Tail Results	Value
Tail $c(y)$	$c(y) = 6 \text{ m} - 0.4296y$
Sref Tail	74.48 m^2
$\bar{c}$ Tail	4.33 m





Center of Gravity Analysis

The table to the below shows the CoG of each loading condition. For AFT and FWD CoG, we have:

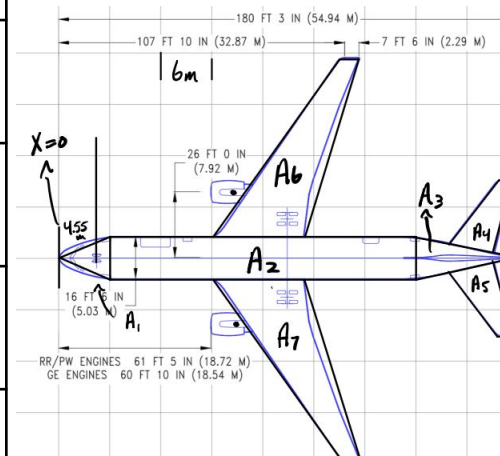
$$X_{cg, AFT} = 25.991m \text{ (105.2\% } X_{cg, aft}/\bar{c}\text{)}$$

$$X_{cg, FWD} = 24.52 m \text{ (85.8\% } X_{cg, FWD}/\bar{c}\text{)}$$

$$x_{cent} = \frac{W_1 \cdot x_1 + W_2 \cdot x_2 + W_3 \cdot x_3 + \dots W_n \cdot x_n}{W_1 + W_2 + W_3 + \dots W_n}$$

Component	Centorid (Measured from the Nose)
Fuselage (A_1 A_2 A_3)	24 m
Wing (Main) (A_6 A_7)	27.29 m
Wing (Tail) (A_4 A_5)	56 m
Engine	21 m
Fuel (Maximum)	27.29 m
Fuel (Minimum)	27.29 m
Passengers/Payload (100kg/Passenger)	26.855 m

Loading Configuration.	Mass (Kg)	Center of Gravity	% of $\bar{c}$
Empty (Fuselage, Wings, Engines, Minimum Fuel)	98,287 kg	24.89 m	90.7%
Max Fuel (Ferrying Flight) (Fuselage, Wings, Engines, Maximum Fuel)	128,287 kg	25.46 m	98.1%
Max Takeoff (Fuselage, Wings, Engines, Maximum Fuel, Payload)	156,987 kg	25.71 m	101.5%
Landing (Fuselage, Wings, Engines, Minimum Fuel, Payload)	126,987 kg	25.33 m	96.6%

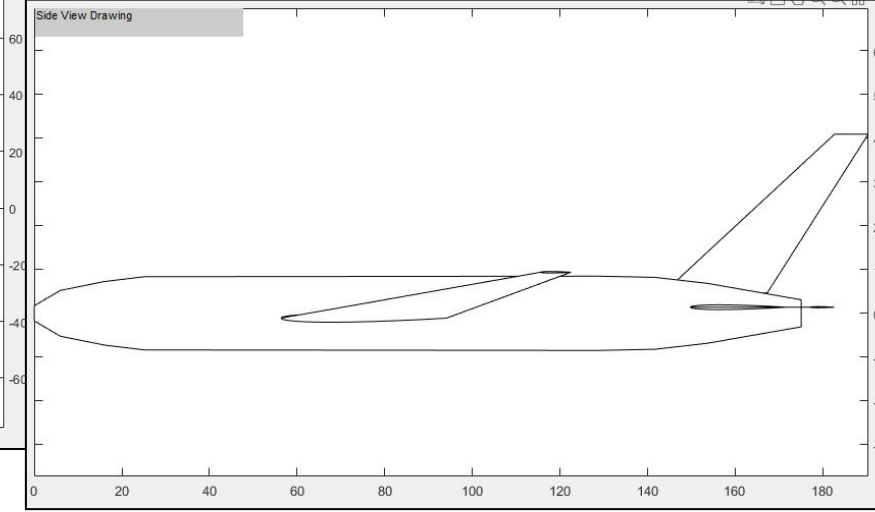
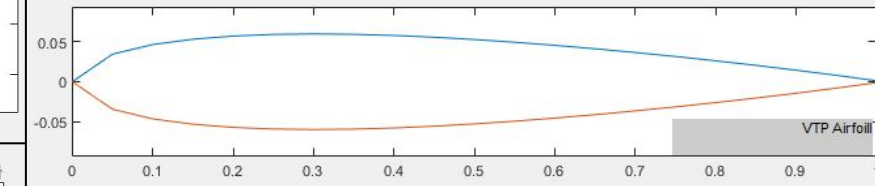
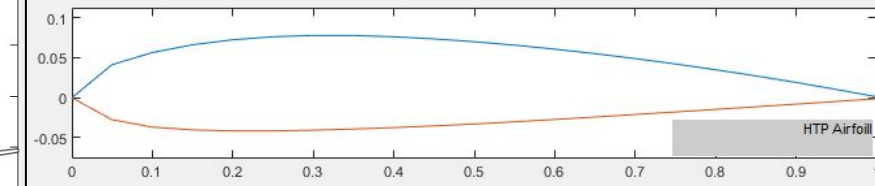
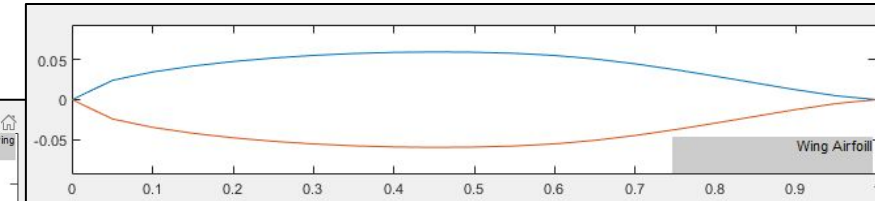
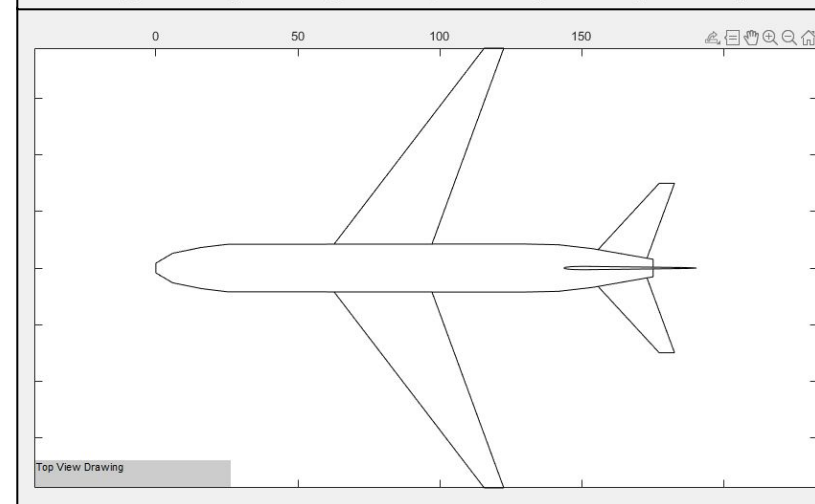
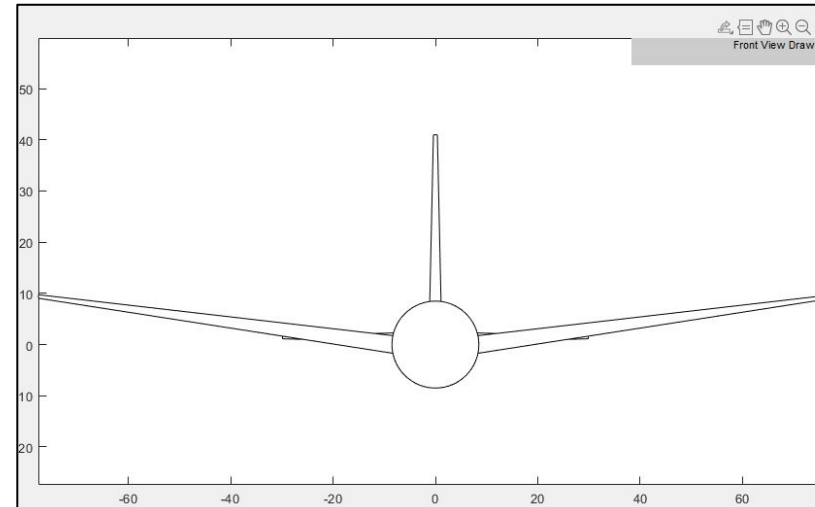


Component	Mass (kg)
Fuselage	14,144 kg
Wing (Main)	6,600 kg (Per Wing)
Wing (Tail)	1,880.5 kg (Per Wing)
Engine	23,591 kg (Per Engine)
Fuel (Maximum)	50,000 kg
Fuel (Minimum)	20,000 kg
Passengers/Payload (100kg/Passenger)	28,700 kg



DATCOM Stability Analysis

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CASEID --
$FLTCN NMACH=2.0,MACH(1)=0.15,0.6,NALPHA=3.0,ALSCHD(1)=-2.0,2.0,
8.0,NALT=1.,ALT(1)=1.0,
WT=345000.0,LOOP=2.$
$SYNTHS XCG=92.2,ZCG=0.0,XW=56.3,ZW=-1.1,ALIW=1.0,XH=149.7,
ZH=1.4,ALIH=0.0,XV=143.6,ZV=4.67$
$OPTINS SREF=3050.0$
$BODY NX=8.0,
X(1)=0.0,5.92,15.98,25.19,128.74,141.71,153.62,175.06,
S(1)=9.40,85.73,165.79,220.28,226.54,212.19,147.40,30.,$
NACA-W-6-66-012
$WGPLNF CHRDP=6.9,SSPNOP=53.2,SSPNE=69.4,SSPN=77.6,
CHRDBP=24.3,CHDR=37.8,SAVSI=33.6,SAVSO=33.6,CHSTAT=0.25,
TWISTA=-3.,DHDADI=7.7,DHDADO=6.,TYPE=1.0$
NACA-H-4-2412-25
$HTPLNF CHRDP=5.4,SSPNE=27.9,SSPN=29.9,CHDR=21.9,SAVSI=38.,
CHSTAT=.25,TYPE=1.0$
NACA-V-4-0012-25
$VTPLNF CHRDP=7.7,SSPNE=31.4,SSPN=36.3,CHDR=23.7,SAVSI=44.0,
CHSTAT=0.25,TYPE=1.0$
DIM FT
BUILD
PLOT
NEXT CASE
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DATCOM Stability Analysis



Oregon State University
College of Engineering

1	AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP WING-BODY-VERTICAL TAIL-HORIZONTAL TAIL CONFIGURATION ---											
FLIGHT CONDITIONS						REFERENCE DIMENSIONS						
MACH NUMBER	ALTITUDE	VELOCITY	PRESSURE		TEMPERATURE	REYNOLDS NUMBER	REF. AREA	REFERENCE LENGTH	MOMENT	REF. CENTER		
	FT	FT/SEC	LB/FT**2		DEG R	1/FT	FT**2	LONG. FT	LAT. FT	HORIZ FT	VERT FT	
0 0.150	1.00	167.45	2.1161E+03		518.666	1.0602E+06	3050.000	24.052	155.200	92.200	0.000	
0	-----DERIVATIVE (PER DEGREE)-----											
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB	CLB	
0												
-2.0	0.015	-0.147	-0.0643	-0.147	0.010	0.436	8.094E-02	-7.745E-03	-1.195E-02	2.209E-03	-2.592E-03	
2.0	0.016	0.177	-0.0938	0.178	0.010	-0.528	8.172E-02	-9.611E-03			-2.928E-03	
8.0	0.041	0.674	-0.1715	0.673	-0.053	-0.255	8.394E-02	-1.629E-02			-3.439E-03	
0				ALPHA	Q/QINF	EPSLON	D(EPSLON)/D(ALPHA)					
0												
				-2.0	1.000	-0.782	0.343					
				2.0	0.982	0.676	0.356					
				8.0	1.000	2.732	0.343					
0*** VEHICLE WEIGHT = 345000.00 LB.												
0*** LEVEL FLIGHT LIFT COEFFICIENT = 3.39386												

Negative CMA	Stable: Aircraft naturally drift towards equilibrium after pitch disturbance
Positive CNB , Negative CLB	Stable: Aircraft naturally generates restoring forces to resist sideslip
Neutral Point , Static Margin	Stable: 1.27, 0.45
Phugoid Damping Freq. & Ratio	Mach 0.6: 0.067 1/s, 0.059 Mach 0.15: 0.27 1/s, 0.067.



Location of Center of Gravities

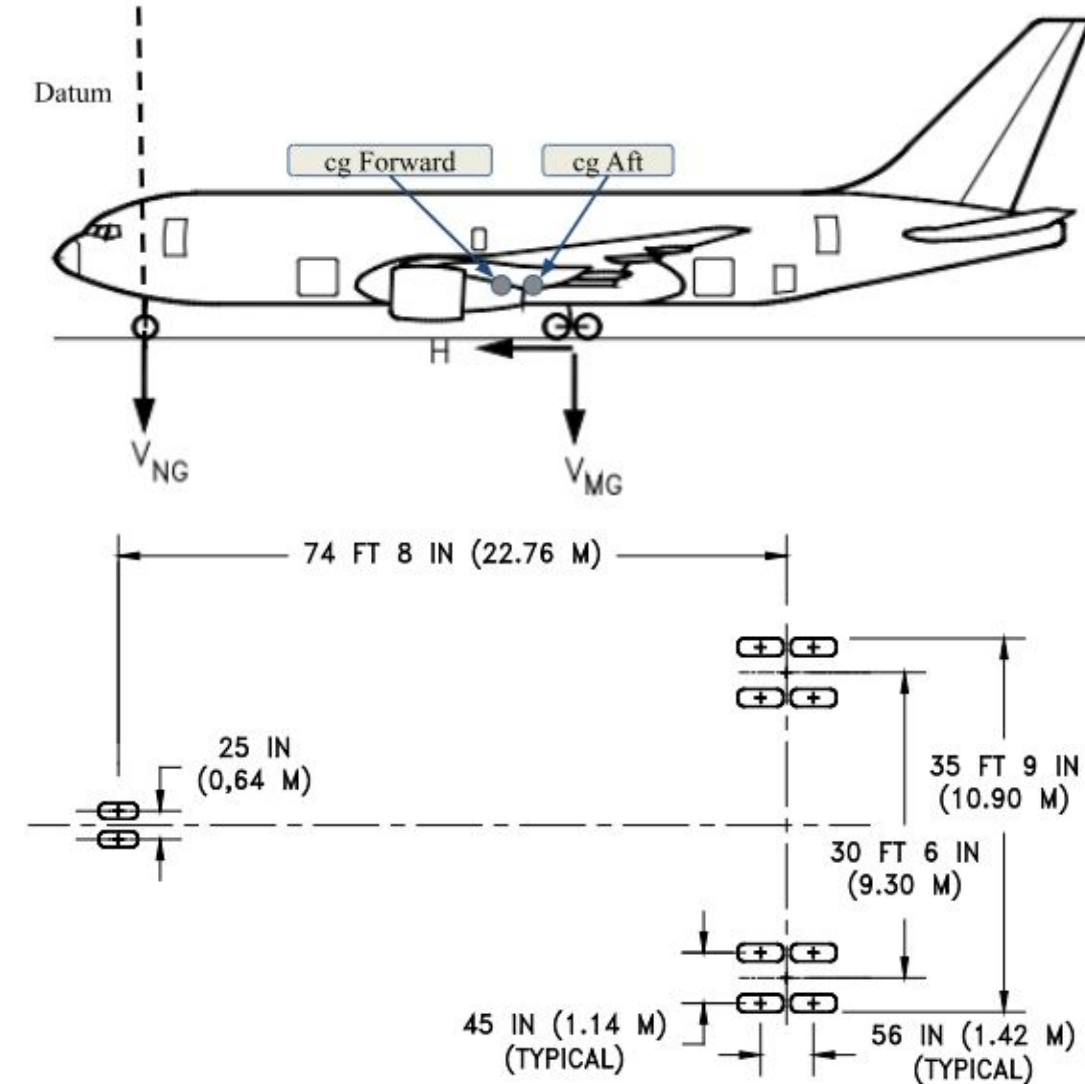
From Boeing's 767 Airplane Characteristics for Airport Planning:

$X_{cg \text{ Aft}} = 21.441 \text{ [m]}$

$X_{cg \text{ Forward}} = 19.916 \text{ [m]}$

$Z_{cg} = 3.947 \text{ [m]}$

W [kg]	163,746
LMG [kg / strut]	77,130
# of struts	2
LNG [kg]	20,455
B [m]	22.76
X_{MLG} [m]	22.76
A [m]	4.65





Landing Gear Loading and Angles

Max LMG	1,513,187.707 [N]
Max LNG	200,440.644 [N]
Min LNG	93,162.553 [N]
Max Breaking LNG	85,189.648 [N]
Tip Back Angle	18.48 degrees
Turn Over Angle	42.606 degrees

Tip Back Angle

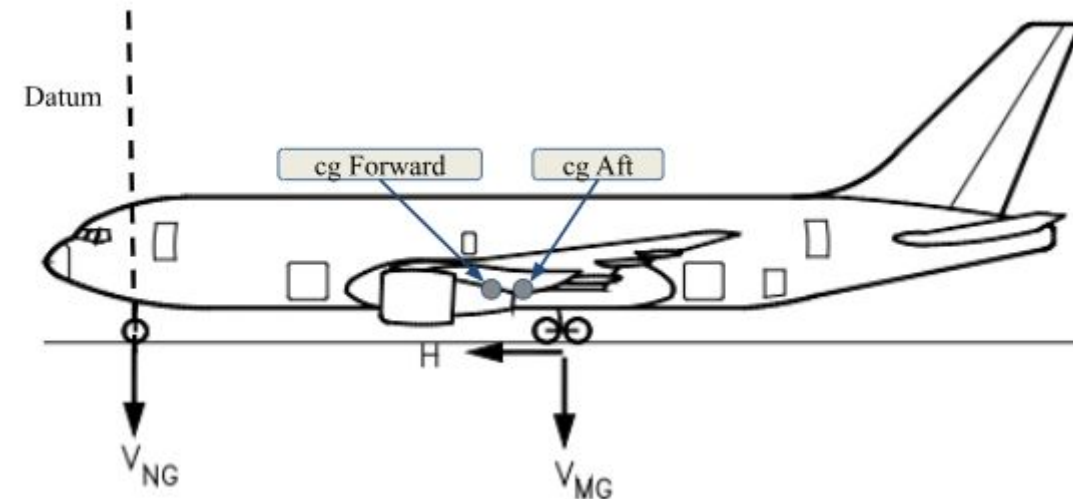
Turn Over Angle

$$\tan \theta = \frac{x_{MLG} - x_{cg \text{ Aft}}}{z_{cg}}$$

$$\alpha = \tan^{-1} \frac{A}{B} \quad \text{where } A = F - C$$

$$X = \frac{C}{\tan \alpha} \quad \theta = \tan^{-1} \frac{H}{Y}$$

$$Y = (D + X) * \sin \alpha \quad H = z_{cg}$$



$$\max L_{MG} = W * \frac{N_a}{B}$$

$$\min L_{NG} = W * \frac{M_a}{B}$$

$$\max L_{NG} = W * \frac{M_f}{B}$$

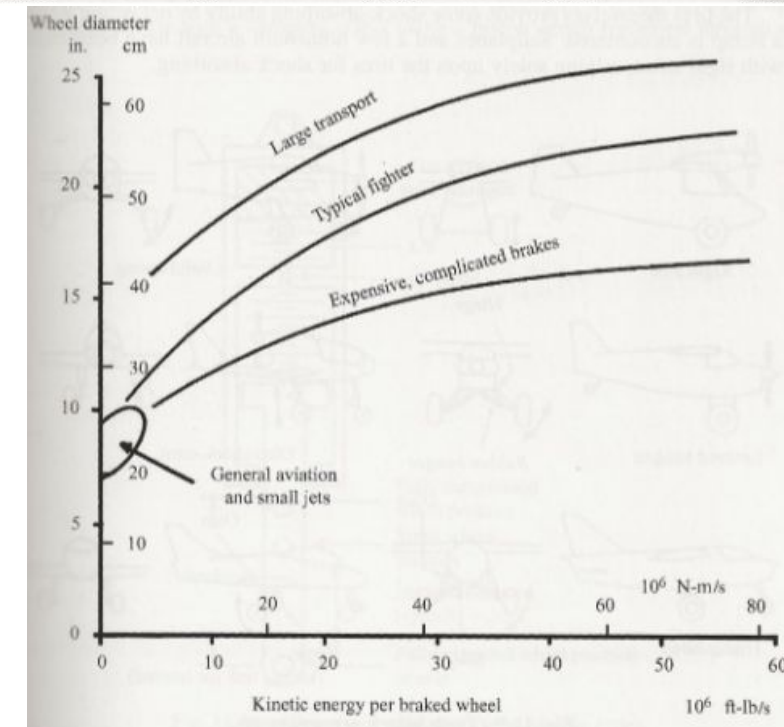
$$\max \text{ braking } L_{NG} = \frac{a}{g} * \frac{H * W}{B}$$



Wheel Size Calculations

# of MLG Wheels	8
# of NLG Wheels	2
W_MLG per Wheel	19,432.50 [kg]
W_NLG per Wheel	10,227.5 [kg]
MLG Tire Size	Diam. = 118.895 [cm] Width = 44.623 [cm]
NLG Tire Size	Diam. = 97.130 [cm] Width = 32.791 [cm]
MLG Tire Size (Kinetic Energy)	Diam. = 60.5 [cm]
NLG Tire Size (Kinetic Energy)	Diam. = 56 [cm]
Actual MLG Size	50x20 R22 32PR
Actual NLG Size	H37x14-15 22PR

	Diameter		Width	
	A	B	A	B
<i>British units: Main wheels diameter or width (in.) = $A W_W^B$</i>				
General aviation	1.51	0.349	0.7150	0.312
Business twin	2.69	0.251	1.170	0.216
Transport/bomber	1.63	0.315	0.1043	0.480
Jet fighter/trainer	1.59	0.302	0.0980	0.467
<i>Metric units: Main wheels diameter or width (cm) = $A W_W^B$</i>				
General aviation	5.1	0.349	2.3	0.312
Business twin	8.3	0.251	3.5	0.216
Transport/bomber	5.3	0.315	0.39	0.480
Jet fighter/trainer	5.1	0.302	0.36	0.467





References

[1] Boeing Commercial Airplanes, “767 Characteristics for Airport Planning” Boeing D6-58328, 2021.

<https://upcommons.upc.edu/bitstream/handle/2117/106436/memoria.pdf;jsessionid=EBDA3CBBF2FE6E385DE1A3E72CB7A53B?sequence=1>

[2] “Boeing 767-300ER Tech Info :: Quick Facts.” Retrieved 9 March 2025.

http://hibdz.skydiving.co.uk/757/767_tech/main/quick_facts.htm

[3] “Boeing 767-300 - Specifications - Technical Data / Description.” Retrieved 9 March 2025.

https://www.flugzeuginfo.net/acdata_php/acdata_7673_en.php

[4] Times, G. B., “How Much Does a Boeing 767 Weigh?,” GB Times, Jul 30 2024. Retrieved 9 March 2025.

<https://gbtimes.com/how-much-does-a-boeing-767-weigh/>